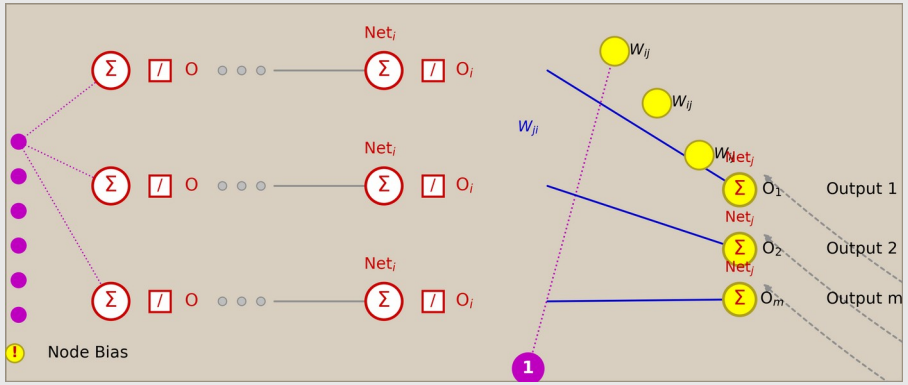
Gradient Descent

$$W_{ji}^{Updated} = W_{ji}^{old} - \eta \frac{\partial E}{\partial W_{ji}}$$

$$W_{ji}^{Updated} = W_{ji}^{old} + \Delta_{ji}$$

$$\text{Where } \Delta_{ji} = -\eta \frac{\partial E}{\partial W_{ji}}$$



Error

Error Class 1

Error Class 2

Error Class m

$$E = \frac{1}{2}\sum_j (t_j - O_j)^2$$

For Sigmoid Activation $O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$

$$Net_j = \sum_i (w_{ji} O_i)$$

$$\frac{\partial E}{\partial O_j} = -(t_j - O_j)$$

$$\frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

$$\frac{\partial Net_j}{\partial W_{ji}} = O_i$$

$$\frac{\partial E}{\partial W_{ji}} = \frac{\partial E}{\partial O_j} * \frac{\partial O_j}{\partial Net_j} * \frac{\partial Net_j}{\partial W_{ji}}$$

$$= -(t_j - O_j) * O_j(1 - O_j) * O_i$$

$$\frac{\partial E}{\partial O_j} = -(t_j - O_j)$$

$$\frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

$$\frac{\partial Net_j}{\partial W_{ji}} = O_i$$